

Chapter-1

Introduction

Introduction

India is predominantly a rural country, with a significant portion of its population engaged in traditional occupations such as handicrafts, weaving, pottery, and other handmade arts. Despite the cultural and economic value of these products, rural artisans often struggle to access wider markets due to geographical limitations, lack of digital literacy, and dependence on intermediaries. With the rapid growth of e-commerce and digital infrastructure in India, there exists a strong opportunity to integrate rural artisans into the digital economy. Graamin E-Haat Bazar is conceptualized to address these challenges by providing a centralized digital marketplace that connects artisans directly with customers.

1.1 Rationale

The Indian rural economy represents approximately 65% of the country's population, yet faces significant digital exclusion and limited market access for traditional crafts and handmade products. Rural artisans, despite producing high-quality handicrafts and textiles, are heavily dependent on: -

Local markets with limited customer reach

Middlemen and wholesalers who take 30-50% profit margins –

Seasonal demand variations

Lack of online presence and digital marketing

1.2 Project Overview

Graamin E-Haat Bazar is a web-based, multi-tier e-commerce platform that:

1. Connects rural artisans directly with urban consumers without middlemen
2. Provides government-verified digital identity for artisans
3. Enables secure online transactions with multiple payment options

4. Facilitates product discovery through intelligent categorization
5. Manages order fulfillment with real-time tracking
6. Ensures quality through rating and feedback mechanisms

Technology Stack:

Frontend: React.js, JavaScript, HTML5/CSS3

Backend: Spring Boot, RESTful API

Database: MySQL with RDBMS normalization

Server: Apache Tomcat

1.3 Objective

The primary objective of the Gramin E-Haat Bazar project is to develop a user-friendly digital marketplace that connects rural artisans directly with urban consumers. The platform aims to eliminate middlemen, thereby ensuring that profits are transferred directly to artisans. Another key objective is to promote rural entrepreneurship while preserving traditional art forms and handicrafts. The system is designed to provide a secure, scalable, and reliable e-commerce infrastructure while implementing modern web technologies to achieve optimal performance. Additionally, the project aligns with government initiatives aimed at promoting digital transformation and inclusive economic development.

1.4 Scope

The scope of the Gramin E-Haat Bazar project includes user registration and authentication for both artisans and customers, product catalog management, and intelligent categorization. It supports shopping cart functionality, checkout processes, order management, and real-time order tracking. The system enables multiple payment options such as UPI, cards, and cash on delivery. It also incorporates a user rating and review mechanism to ensure quality assurance. An admin dashboard is included for system monitoring and management. Furthermore, the project covers database schema design, normalization, ER diagram creation, data dictionary documentation, database

optimization, indexing, security implementation, and comprehensive testing documentation- Test documentation

- Database schema design and normalization
- Entity Relationship Diagram (ERD) creation
- Data dictionary documentation
- Database optimization and indexing

1.5 Roles & Responsibilities

Aditya Singh Bhati (0832CS231013)

- Design and develop RESTful APIs
- Implement Spring Boot backend architecture
- Integrate payment gateways
- Authentication and authorization mechanisms
- Server-side validation and business logic

Akshay Singh Solanki (0832CS231018)

- Design user interface mockups
- Develop React.js components
- Implement responsive design
- User experience optimization
- Frontend form validation and AJAX integration

Dheeraj Dayal (0832CS231059)

- Database schema design and normalization
- Entity Relationship Diagram (ERD) creation
- Data dictionary documentation
- Database optimization and indexing

1.6 Report Organization

Chapter 1: Introduction

This chapter provides an overall introduction to the project by establishing its background and relevance. It explains the context in which the project has been developed, highlighting the need for a structured and efficient solution. The chapter clearly defines the objectives of the project, outlining what the system aims to achieve. It also describes the scope of the project by specifying its boundaries and functionalities. Additionally, this chapter presents the team structure, detailing the roles and responsibilities of team members involved in the development of the project.

Chapter 2: Study of Existing System and Requirements

This chapter focuses on the analysis of existing e-commerce platforms and systems to understand their working models, strengths, and limitations. It identifies gaps and challenges in the current systems, particularly in terms of scalability, accessibility, and transparency. Based on this analysis, the chapter proposes suitable solutions to overcome the identified shortcomings. It also includes a detailed feasibility analysis covering technical, economic, and operational feasibility to evaluate the practicality of the proposed system.

Chapter 3: Project Planning

This chapter outlines the project planning process and provides a structured roadmap for successful project execution. It presents a Gantt chart and detailed project schedule to illustrate task sequencing, timelines, and milestones. The chapter also discusses resource allocation, including human resources, hardware, and software requirements, along with budget estimation. Furthermore, it addresses potential risks associated with the project and explains the risk management strategies adopted to minimize uncertainties and ensure timely completion.

Chapter 4: System Analysis

This chapter presents a comprehensive analysis of the system requirements. It specifies both functional and non-functional requirements, clearly defining system behavior, performance expectations, and constraints. The chapter documents the Software

Requirements Specification (SRS), serving as a formal reference for system development. Additionally, it details interface requirements, including user interfaces, hardware interfaces, and software interfaces, ensuring smooth interaction between system components

Chapter 5: System Analysis Tools

This chapter describes the design tools and techniques used in the development of the system. It presents various UML diagrams such as Data Flow Diagrams (DFD), Use Case Diagrams, and Class Diagrams to visually represent system architecture, data movement, and object-oriented structure. These diagrams help in understanding system functionality, interactions, and design logic, thereby serving as a blueprint for implementation

Chapter 6: Software Test Documentation

This chapter outlines the testing strategy adopted to ensure the quality and reliability of the software. It explains the test plans and testing methodologies used during different stages of development. The chapter documents detailed test cases for all system modules, ensuring comprehensive validation of functionality.

Chapter 7: Software Test Documentation

This chapter summarizes the overall achievements of the project, highlighting how the stated objectives have been successfully met. It evaluates the effectiveness of the developed system in addressing the identified problems. The chapter also discusses the future scope of the project by suggesting possible enhancements and extensions, such as the integration of advanced technologies, scalability improvements, and additional features that can further improve system functionality and usability

Chapter-2

Study of Existing System and System Requirements

2.1 Identification of Need

2.1.1 Existing System & its drawbacks

1. Traditional Haat Bazaar Model

- Physical location-based marketplace
- Limited to local customer base
- Time and logistics constraints
- Seasonal variations in foot traffic
- No digital presence

2. Existing E-Commerce Platforms (Amazon, Flipkart)

- Complex onboarding for rural sellers
- High commission rates (15-30%)
- Difficult payment settlement processes
- Lack of artisan-specific features
- Require bulk stock and warehouse management

3. Government Initiatives (MP Handicraft Portal)

- Limited product categories
- Not a real-time marketplace
- Manual intervention required
- Low transaction volumes
- Poor user interface for rural users

2.2 Preliminary Investigation

2.2.1 Proposed System & its advantages

The proposed system, Gramin E-Haat Bazar, is designed as an artisan-centric digital marketplace that prioritizes ease of use, accessibility, and fairness. The platform features a simple and intuitive user interface specifically designed for rural users with minimal digital literacy. To enhance inclusivity, the system includes a roadmap for localized language support, allowing artisans to interact with the platform in their preferred regional languages.

One of the key advantages of the proposed system is direct market access for artisans. The platform enables pan-India customer reach by providing a 24/7 online storefront, allowing artisans to showcase and sell their products at any time. Real-time product updates ensure that inventory and pricing remain accurate, while the digital nature of the platform offers global visibility potential, expanding market opportunities beyond geographical limitations.

The system also incorporates a fair pricing model that ensures economic transparency and sustainability. A low commission structure ranging between 5% and 10% significantly reduces financial burden on artisans compared to traditional intermediaries. The model supports direct profit transfer to artisans, ensuring timely and secure payments.

Transparent pricing mechanisms and the absence of hidden charges build trust among both artisans and customers, promoting long-term platform adoption

2.3 Feasibility Study

The feasibility study evaluates the practicality of the proposed system by analyzing technical, economic, and operational aspects. This study ensures that the system can be successfully developed, implemented, and maintained within the available resources and constraints

2.3.1 Technical Feasibility

The proposed system is technically feasible due to the availability of reliable, mature, and well-supported technologies. Spring Boot is used as the backend framework, offering

robustness, scalability, and extensive documentation, which facilitates rapid development and maintenance. React.js is selected for frontend development due to its component-based architecture, responsiveness, and widespread industry adoption. MySQL serves as the database management system, providing reliable data storage, efficient query handling, and proven scalability for large datasets. Additionally, cloud infrastructure platforms such as AWS and Azure are available for deployment, offering scalability, security, and high availability, making the system suitable for future expansion

Available Technologies:

- Spring Boot: Mature, well-documented framework
- React.js: Industry-standard frontend framework
- MySQL: Reliable RDBMS with proven scalability
- Cloud Infrastructure: AWS/Azure available for deploy

2.3.3 Economical Feasibility

Cost Analysis:

Component	Cost (INR)	Remarks
Development Tools	0	Open-source (Spring Boot, React, MySQL)
Cloud Hosting (6 months)	15,000	AWS/Azure free tier + minimal paid
Domain & SSL	2,000	Annual certificate and domain
Testing Tools	0	Open-source (Selenium, JUnit)
Documentation	0	Internal using free tools
Total Initial Cost	17,000	Minimal investment

Revenue Model (Post-Launch):

- Transaction commission: 5-10% per order
- Advertising revenue: Featured listings
- Premium seller packages: Enhanced visibility
- Logistics integration revenue: Commission on shipping

- **Identified Drawbacks:**

Issue	Current System	Impact
Market Access	Limited to local area	Reduced sales potential
Middlemen	30-50% profit loss	Lower artisan income
Digital Literacy	Complex platforms	Adoption barriers
Payment Methods	Limited options	Customer inconvenience
Verification	Manual and time-consuming	Delayed activation

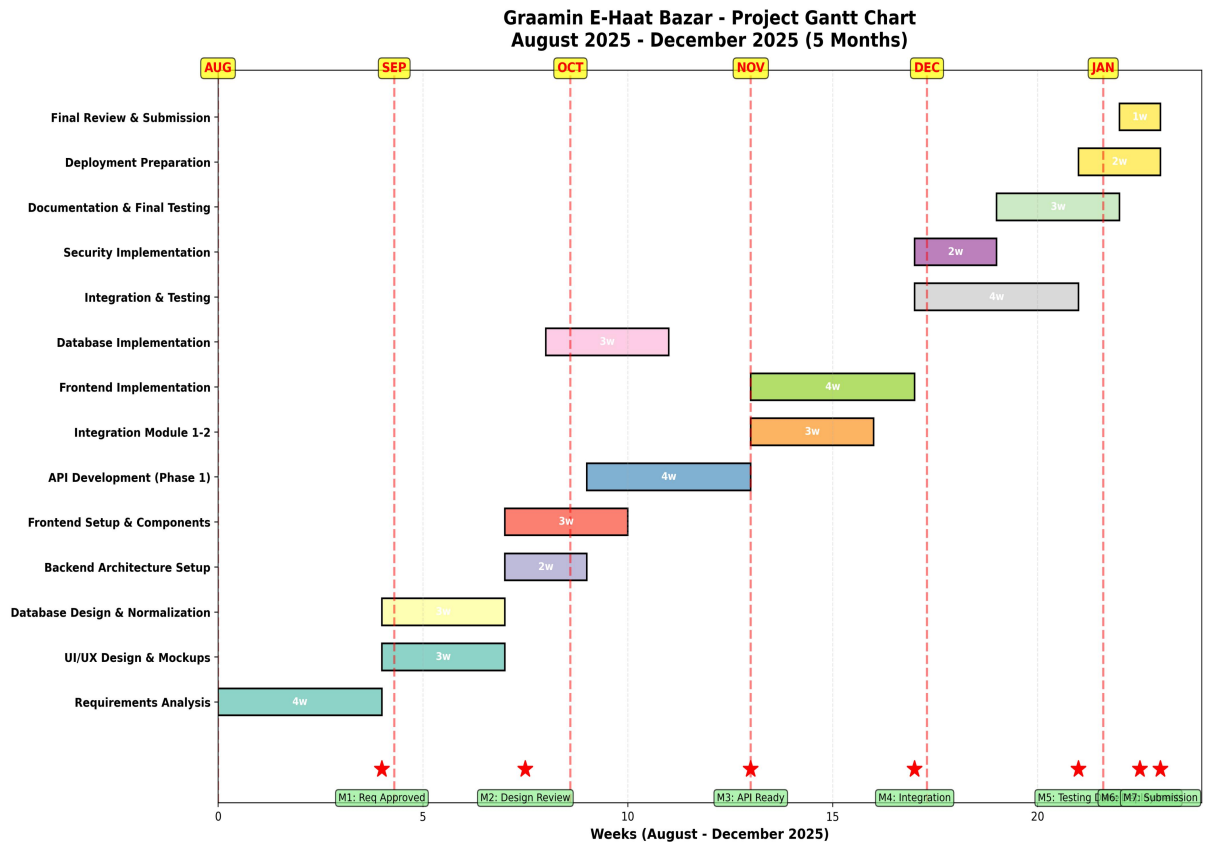
Chapter-3

Project Planning

3.1 Schedule & Milestone

3.1.1 Gantt Chart

A Gantt chart is a widely used project management tool that provides a visual representation of a project schedule over time. It illustrates the sequence of project activities, their start and end dates, and their duration in a bar chart format. By presenting tasks along a timeline, the Gantt chart helps in effective planning, monitoring, and control of project progress. It enables project managers and team members to understand task dependencies and track milestones throughout the project lifecycle.



3.1.2 Project Schedule Table

Phase	Activity	Start Date	End Date	Duration	Responsible	Status
Phase 1: Planning	Requirements Analysis	Aug 1	Aug 30	4 weeks	All	Completed
	Document SRS	Aug 15	Aug 30	2 weeks	Aditya	Completed
Phase 2: Design	UI/UX Mockups	Sep 1	Sep 20	3 weeks	Akshay	In Progress
	Database ERD	Sep 1	Sep 15	2 weeks	Dheeraj	In Progress
	System Architecture	Sep 10	Sep 25	2 weeks	Aditya	In Progress
Phase 3: Development	Backend API Development	Oct 1	Oct 30	4 weeks	Aditya	Scheduled
	Frontend Component Dev	Oct 1	Oct 30	4 weeks	Akshay	Scheduled
	Database Implementation	Sep 20	Oct 15	3 weeks	Dheeraj	Scheduled
Phase 4: Integration	API & Frontend Integration	Nov 1	Nov 15	2 weeks	Aditya, Akshay	Scheduled
	Database Integration	Nov 1	Nov 10	1 week	Dheeraj	Scheduled
Phase 5: Testing	Unit Testing	Nov 5	Nov 15	1 week	All	Scheduled

Project Schedule Table

3.2 Resources and Budget

Resource	Quantity	Unit Cost	Total Cost	Notes
Hardware				
Development Laptops	3	Owned	0	Team members' laptops
Testing Devices	2	Owned	0	Smartphones/Tablets
Software				
IDE (IntelliJ, VS Code)	3	Free	0	Community edition
Git Repository (GitHub)	1	Free	0	Free tier
Hosting & Deployment				
Cloud Server (AWS/Azure)	6 months	2,500/month	15,000	Free tier + minimal
Domain Name	1	2,000	2,000	Annual
Documentation				
Tools(Markdown, Confluence)	N/A	Free	0	Open source
Printing & Binding	5 copies	500	2,500	Final report
Testing				
Manual Testing Labor	60 hours	0	0	Team effort
TOTAL PROJECT COST				19,500 INR

Resource Allocation

3.3 Risk Management

Risk	Probability	Impact	Severity	Mitigation Strategy
Technical Integration Issues	Medium	High	High	Early integration testing, weekly sync
Database Performance	Low	High	Medium	Indexing, optimization from start
Payment Gateway Integration	Low	High	Medium	Use sandbox for testing, partner liaison
Security Vulnerabilities	Low	Critical	High	Security audit, penetration testing
Team Member Illness	Low	Medium	Low	Knowledge sharing, documentation
Scope Creep	Medium	Medium	Medium	Strict requirement freezing
API Design Changes	Medium	Medium	Medium	Versioning strategy, backward compatibility
Cloud Deployment Issues	Low	Medium	Low	Infrastructure as code, testing
Timeline Slippage	Medium	Medium	Medium	Buffer time, priority-based phasing
Data Loss	Very Low	Critical	High	Regular backups, version control

Risk Identification and Mitigation

Chapter 4:

System Analysis

4.1 Software Requirement Specification

The Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements of the system, including user registration, product listing, shopping cart, online payment, order management, and delivery tracking features. It also specifies system requirements related to security, performance, usability, and reliability to ensure a smooth and secure online shopping experience. The SRS serves as a foundational document that guides the design, development, and testing of the e-commerce platform.

4.1.1 External Interface Requirement

The external interface requirements describe how the Gramin E-Haat Bazar e-commerce system interacts with users and external systems. The user interface is web-based and designed to be simple, responsive, and user-friendly, enabling artisans, customers, and administrators to access the system through standard web browsers. The system interfaces with external payment gateways to support secure online transactions such as UPI, debit cards, credit cards, and cash on delivery. It also interacts with external services for order tracking and notification purposes. These interfaces ensure seamless communication between the system and external components while maintaining security and reliability.

4.1.1.1 User Interface

The user interface of the Gramin E-Haat Bazar e-commerce website is designed to be intuitive, responsive, and user-friendly, catering to customers, artisans, and administrators. The **Customer Module UI** provides a personalized dashboard displaying recommended products based on user preferences. It includes advanced search functionality with filters such as price, category, and ratings to help users find products easily. Detailed product

pages display images, specifications, and pricing information. Customers can manage items in the shopping cart, proceed to checkout by selecting delivery addresses and payment options, view order history, track orders in real time, manage their profiles, and submit ratings and reviews for purchased products.

The **Artisan Module UI** focuses on simplicity and ease of use to accommodate users with limited digital experience. It allows artisans to register and authenticate securely, manage their product listings by adding, editing, or deleting items, and track inventory levels. Artisans can view orders through a dedicated dashboard and communicate with customers when required, ensuring smooth order fulfillment and customer satisfaction.

The **Admin Module UI** is designed for effective system management and control. It enables administrators to manage and verify users, moderate product listings to maintain quality standards, oversee order processing, and monitor transactions. This interface ensures smooth platform operation, security, and compliance with system policies.

4.1.1.2 Hardware Interface

The Gramin E-Haat Bazar e-commerce system has specific hardware and software requirements to ensure smooth operation and optimal performance. On the **server side**, the system requires a processor of at least dual-core or Intel i5 and above, with a minimum of 8 GB RAM to efficiently handle concurrent user requests and background processes. Storage should be at least 500 GB SSD to store the application, database, and transaction data. A reliable network connection of 100 Mbps Ethernet is recommended to ensure fast and uninterrupted access for users.

On the **client side**, customers and artisans can access the platform using desktop computers or laptops with modern web browsers such as Chrome, Firefox, Safari, or Edge. For mobile access, the system supports devices running Android 8.0 and above or iOS 12.0 and above, ensuring compatibility with a wide range of smartphones. The display should have a minimum width of 320 pixels to maintain responsive layouts and proper visualization of the user interface. These system requirements collectively ensure

that the platform operates efficiently, providing a seamless and reliable experience for all users

On the server side, the application is deployed on reliable hardware capable of supporting web-based services, including application servers and database servers. These servers must have sufficient processing power, memory, and storage to handle concurrent user requests, transaction processing, and data storage. The hardware interface is designed to be scalable, allowing upgrades as the number of users and transactions increases. This flexibility ensures consistent system performance, reliability, and uninterrupted service delivery.

4.1.1.3 Communication Interface

The Gramin E-Haat Bazar system uses standard communication protocols to ensure secure and reliable data exchange between clients and the server. All interactions are carried out over HTTP or HTTPS with TLS 1.2 or higher to protect sensitive information. The system provides RESTful API endpoints to handle all core operations, including user management, product listing, orders, and payments. Additionally, webhook support is implemented for real-time payment notifications, enabling the system to update order status immediately after transaction confirmation. This communication interface ensures seamless, secure, and efficient connectivity across all components of the platform

The system relies on RESTful API endpoints to perform all essential operations, including user registration, authentication, product management, order processing, and payment handling. These APIs provide a standardized interface for interaction between the frontend and backend components, enabling modularity, scalability, and maintainability of the system. In addition, the platform supports webhook integration for real-time notifications from payment gateways, ensuring that order statuses are updated instantly after successful or failed transactions. This feature enhances operational efficiency by allowing immediate feedback to both customers and artisans

4.2 Functional Requirements

• Customer Module

FR-C01	User Registration	Customer can register with email, password, and basic details
FR-C02	User Login	Secure authentication with email/password
FR-C03	Product Search	Search products by name, category, price range
FR-C04	Product Browsing	View product details, images, ratings, reviews
FR-C05	Shopping Cart	Add/remove items, update quantities
FR-C06	Checkout	Enter shipping address, select payment method
FR-C07	Order Placement	Confirm order with payment
FR-C08	Order Tracking	View order status and location updates
FR-C09	Rating & Review	Submit ratings and feedback on products
FR-C10	Profile Management	Update address, password, preferences
FR-C11	Order History	View past orders and reorder functionality

• Artisan Module

FR-A01	Registration with Verification	Register with Aadhaar/Government ID verification
FR-A02	Product Management	Add, edit, delete products with images

FR-A03	Inventory Management	Update stock levels and availability
FR-A04	Order Acceptance	View and accept/reject customer orders
FR-A05	Order Fulfillment	Mark orders as packed, shipped
FR-A06	Customer Communication	Message customers regarding orders
FR-A07	Earnings Dashboard	Track sales, revenue, commissions
FR-A08	Profile Management	Update skills, experience, shop details

• Admin Module

FR-AD01	User Management	View, verify, approve/reject user registrations
FR-AD02	Product Moderation	Review and approve/reject product listings
FR-AD03	Category Management	Add/edit/delete product categories
FR-AD04	Order Monitoring	Track all orders and transactions
FR-AD05	Payment Settlement	Process payments to artisans
FR-AD06	Complaint Resolution	Handle disputes and complaints
FR-AD07	Analytics	System-wide analytics and reporting
FR-AD08	Content Management	Manage banners, promotions, announcements
FR-AD09	Commission Management	Set and modify commission rates
FR-AD10	Audit Logs	Maintain activity logs for compliance

4.3 Non-Functional Requirements

Reliability

- System availability: 99.5% uptime
- Automatic backup every 6 hours
- Disaster recovery plan within 24 hours
- Error handling and logging

Availability

- 24/7 system operation

- No scheduled downtime except maintenance
- Graceful degradation during issues
- Load balancing for high traffic

Security

- SSL/TLS encryption for data in transit
- Password hashing (bcrypt)
- SQL injection prevention (Prepared statements)
- XSS and CSRF protection
- Role-based access control (RBAC)
- PCI DSS compliance for payments

Maintainability

- Modular code structure
- Comprehensive code documentation
- Database schema normalization
- Version control (Git)
- Automated testing

Portability

- Database independent queries
- Compatible with major browsers
- Cloud-ready architecture
- API-first design

Performance

- Page load time < 3 sec

- API response time < 500ms (95th percentile)
- Support for 10,000 concurrent users
- Database query optimization
- Client-side caching

4.4 My Contribution in System Analysis

Aditya Singh Bhati

- Defined functional requirements for all modules
- Designed REST API specifications
- Documented security requirements
- Created SRS document
- Defined external interface requirements

Akshay Singh Solanki

- Designed UI/UX requirements
- Documented user interaction flows
- Specified web browser compatibility
- Created responsive design specifications
- Defined accessibility requirements

Dheeraj Dayal

- Analyzed database requirements
- Designed data structure requirements
- Defined database performance requirements

Specified data integrity constraints

Chapter 5:

System Analysis Tools

5.1 Introduction

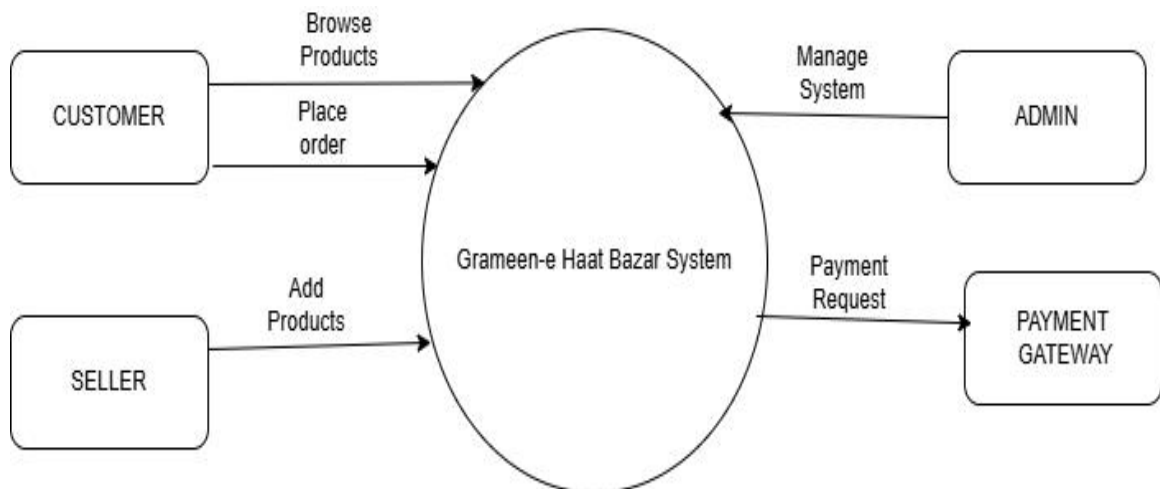
This chapter documents the system design using UML (Unified Modeling Language) and database design tools. The design phase translates system requirements into architectural specifications that guide implementation

5.2 Design Overview

5.2.1 Data Flow Diagram (DFD)

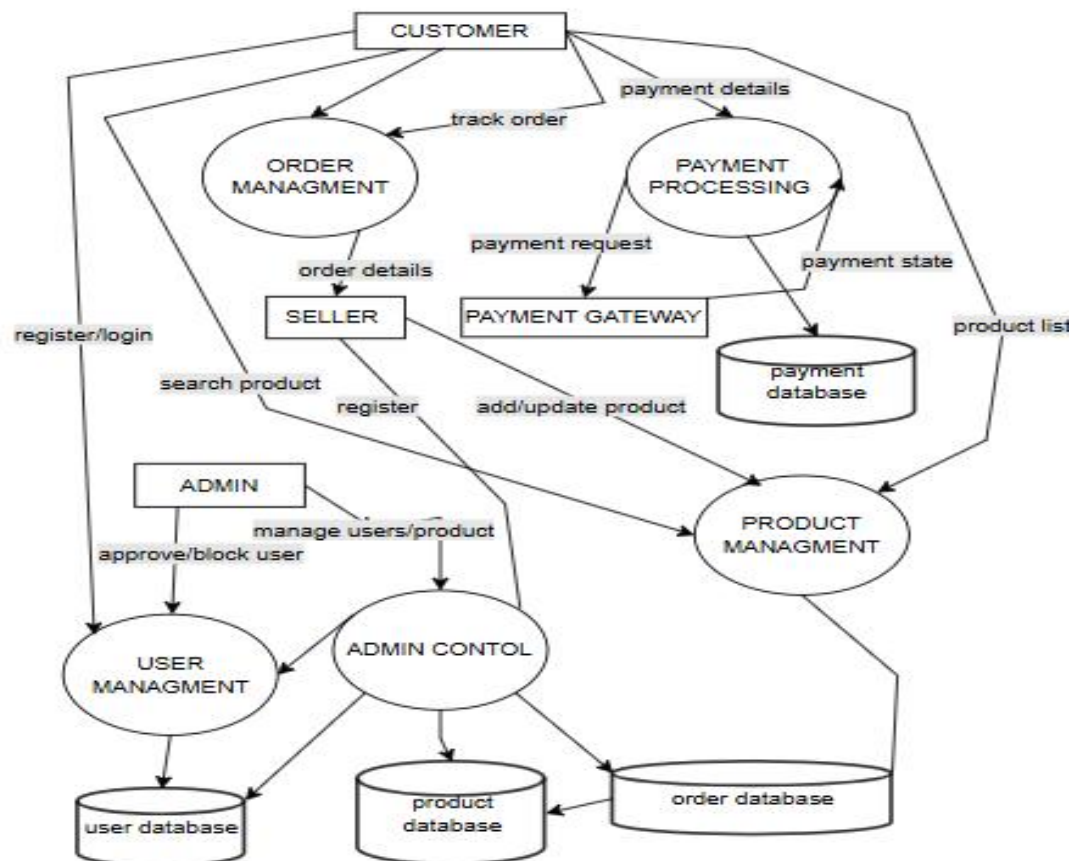
Level 0 DFD (Context Diagram)

The system receives product queries, order placements, and payments from customers. Artisans provide product updates and order information, while admins monitor the system. The core system processes all these inputs and maintains a database.



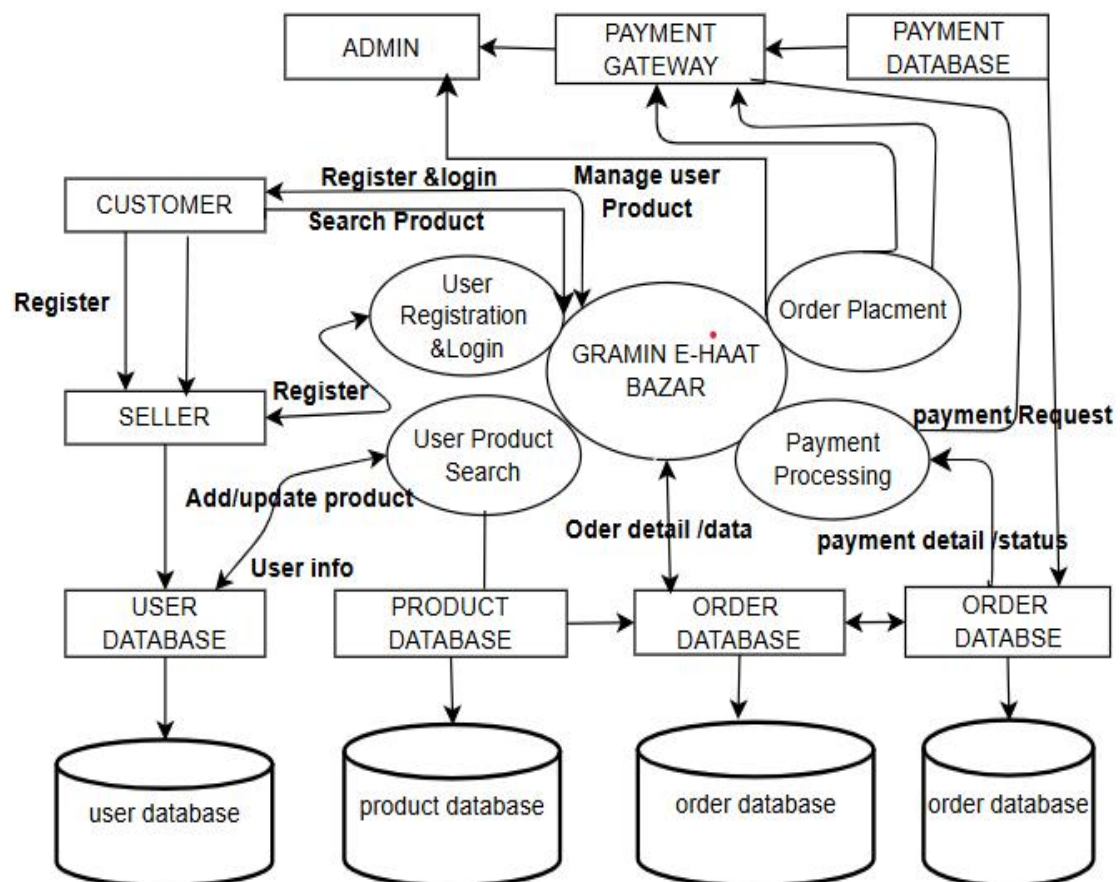
Level 1 DFD (Major Processes)

The system is decomposed into major functional modules to show the internal flow of data. These modules include User Management, Product Management, Order Management, Payment Processing, Delivery and Tracking, Review and Rating Management, and Admin Management. Each module interacts with the central database to store or retrieve relevant information. For instance, User Management handles authentication and profile updates, Product Management allows artisans to manage product details, and Order Management coordinates with Payment Processing and Delivery modules to ensure accurate and timely order fulfillment. Review and Rating Management captures customer feedback, while Admin Management oversees overall system control and monitoring. The Level 1 DFD provides a detailed representation of how data moves between modules and external entities, illustrating the functional architecture of the e-commerce platform in a clear and structured manner.



Level 2 DFD (Major Processes)

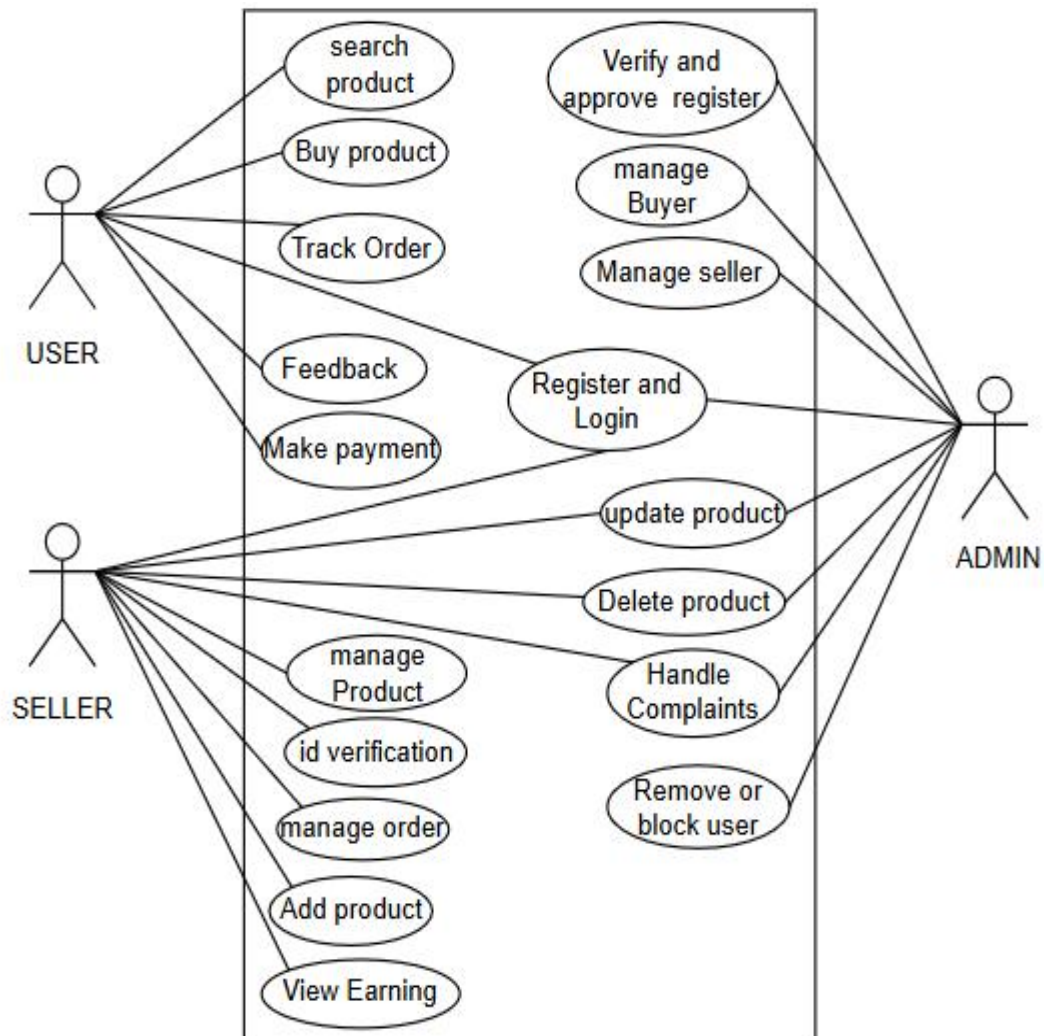
The Level 2 Data Flow Diagram (DFD) of Gramin E-Haat Bazar provides a more detailed breakdown of specific modules from the Level 1 DFD, illustrating the step-by-step data flow within individual processes. For example, in the **Order Management** module, Level 2 DFD shows sub-processes such as order placement, payment verification, inventory update, and order confirmation. When a customer places an order, the system verifies product availability and sends the payment request to the Payment Processing module. Once the payment is confirmed, the inventory is updated, and an order confirmation is sent to the customer. Simultaneously, delivery information is forwarded to the Delivery and Tracking module, which generates shipping updates and tracks order status in real time.



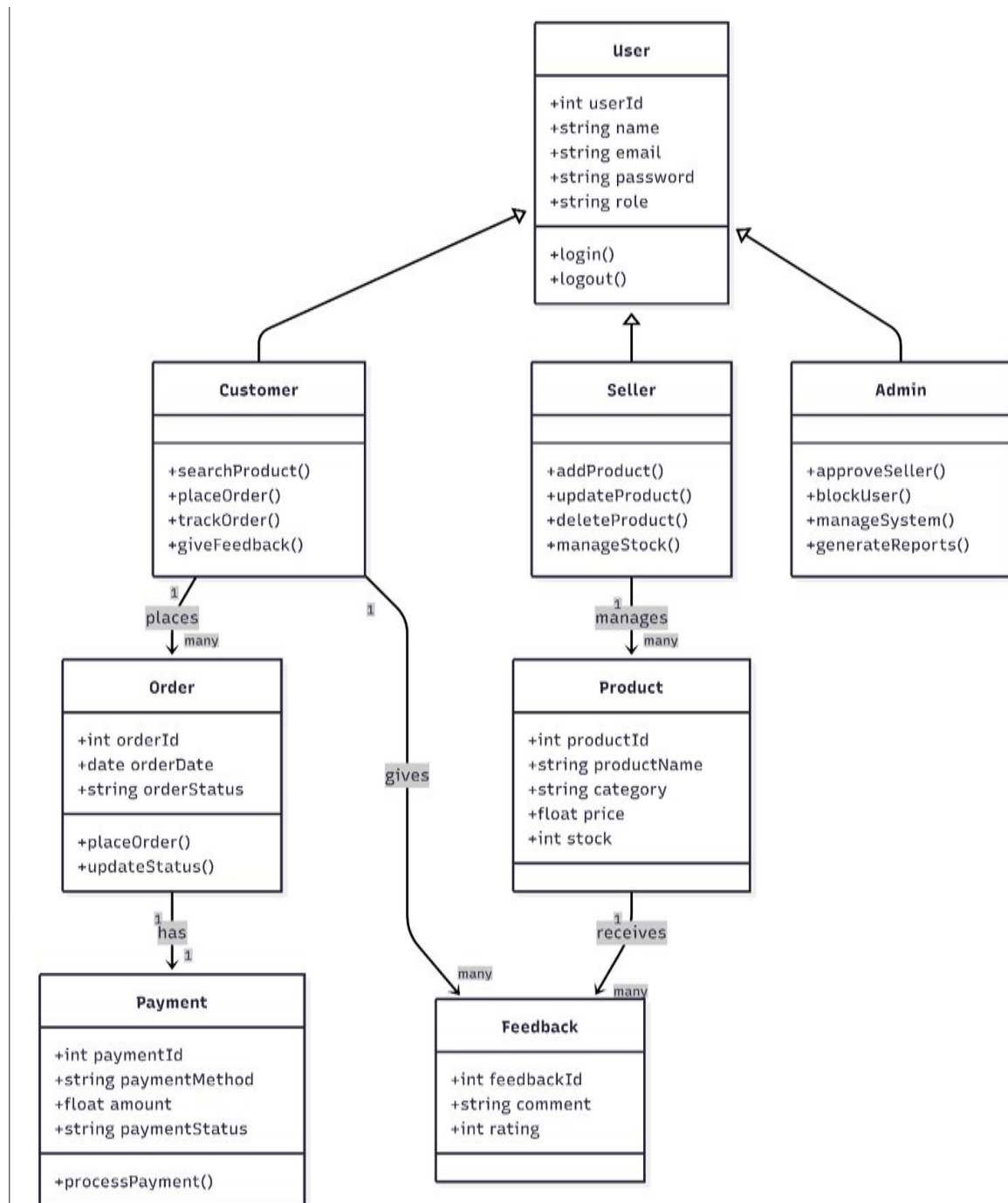
2 Use Case Diagram

Actor Analysis:

- Customers: Can register, search, purchase, track orders, and rate products
- Artisans: Can register with verification, manage products, track earnings, and communicate with customers
- Admin: Can verify users, moderate content, settle payments, and manage the system
- Payment Gateway: Processes payments and sends confirmations



5.2.3 Class Diagram



5.3.1 Conversion of E-R Diagram into Tables

ADMIN TABLE

admin_id(PK)	password

USER TABLE

User_id(PK)	email	name	password

PAYMENT TABLE

Payment_id(PK)	User_id	amount	status	Oder_item_id	method

PRODUCT TABLE

Product_id(PK)	name	price	stock	description	Category_id

CATEGORY TABLE

Category_id(PK)	Category name

ODER TABLE

Oder_id(PK)	Uder_id	Order date	Total amount

ORDER_ITEM_TABLE

Order_item_id(PK)	Oder_id	Product_id	quantity

TRACKING TABLE

Tracking_id(PK)	Order_id	Status

DELIVERY TABLE

Delivery_id(PK)	Delivery date	Oder_id	Expected date	status

ADDRESS TABLE

Address_id	city	state	Pin code	User_id

5.3.2 Normalization Details

Table	Status	Atomic Values	No Repeating Groups	Primary Key
USER	✓	All columns contain single values	No multivalued attributes	user_id
ADDRESS	✓	All columns contain single values	No repeating address elements	address_id
PRODUCT	✓	All columns contain single values	No repeating product attributes	product_id
CATEGORY	✓	All columns contain single values	Single category per record	category_id
ORDER	✓	All columns contain single values	No repeating order details	order_id
ORDER_ITEM	✓	All columns contain single values	Single item per record	order_item_id
PAYMENT	✓	All columns contain single values	One payment per record	payment_id

First Normal Form (1NF) Compliance

Table	Status	Composite Key	Full Functional Dependency	Partial Dependency
USER	✓	No	N/A	None
ADDRESS	✓	No	N/A	None
PRODUCT	✓	No	N/A	None
CATEGORY	✓	No	N/A	None
ORDER	✓	No	N/A	None
ORDER_ITEM	✓	Yes (order_id + prod_id)	All attributes fully dependent	None
PAYMENT	✓	No	N/A	None
TRACKING	✓	No	N/A	None
RATING	✓	No	N/A	None

Second Normal Form (2NF) Compliance

Table Name	Primary Key(s)	3NF Status	Explanation
USER	user_id	✓ Compliant	All attributes depend on user_id
ADDRESS	address_id	✓ Compliant	All attributes depend on address_id
PRODUCT	product_id	✓ Compliant	All attributes depend on product_id
CATEGORY	category_id	✓ Compliant	All attributes depend on category_id
ORDER	order_id	✓ Compliant	All attributes depend on order_id
ORDER_ITEM	(order_id, product_id)	✓ Compliant	All attributes depend on composite key
PAYMENT	payment_id	✓ Compliant	All attributes depend on payment_id
TRACKING	tracking_id	✓ Compliant	All attributes depend on tracking_id
RATING	rating_id	✓ Compliant	All attributes depend on rating_id (see the generated image above)

Third Normal Form (3NF) Compliance

5.3.3 Table Structure

Column	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
name	VARCHAR(100)	NOT NULL	User's full name
email	VARCHAR(100)	UNIQUE, NOT NULL	Email for login and contact
password	VARCHAR(255)	NOT NULL	Encrypted password (bcrypt)
phone	VARCHAR(15)	NOT NULL	Contact phone number
user_type	ENUM('CUSTOMER','ARTISAN','ADMIN')	NOT NULL	Type of user
is_verified	BOOLEAN	DEFAULT FALSE	Account verification status

USER Table

Column	Data Type	Constraints	Description
product_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique product identifier
artisan_id	INT	NOT NULL, FOREIGN KEY	Reference to USER table
category_id	INT	NOT NULL, FOREIGN KEY	Reference to CATEGORY
product_name	VARCHAR(100)	NOT NULL	Product title
description	TEXT	NOT NULL	Detailed product description
price	DECIMAL(10,2)	NOT NULL	Selling price per unit
stock_quantity	INT	NOT NULL, DEFAULT 0	Current inventory
image_url	VARCHAR(255)	NULL	Product photo URL
status	ENUM('ACTIVE','INACTIVE','PENDING')	DEFAULT 'PENDING'	Product status

PRODUCT Table

Column	Data Type	Constraints	Description
order_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique order number
customer_id	INT	NOT NULL, FOREIGN KEY	Reference to CUSTOMER (USER)
shipping_address_id	INT	NOT NULL, FOREIGN KEY	Reference to ADDRESS
order_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Order placement timestamp
total_amount	DECIMAL(10,2)	NOT NULL	Total order value
order_status	ENUM('PENDING','CONFIRMED','SHIPPED','DELIVERED','CANCELLED')	DEFAULT 'PENDING'	Order status
payment_method	ENUM('UPI','CREDIT_CARD','DEBIT_CARD','CREDIT')	NOT NULL	Payment method used

ORDER Table

Column	Data Type	Constraints	Description
payment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique payment transaction ID
order_id	INT	NOT NULL, UNIQUE, FOREIGN KEY	Reference to ORDER
user_id	INT	NOT NULL, FOREIGN KEY	Reference to USER (payer)
amount	DECIMAL(10,2)	NOT NULL	Amount paid
payment_method	VARCHAR(50)	NOT NULL	Payment method used
transaction_id	VARCHAR(100)	NOT NULL	Gateway transaction reference
payment_status	ENUM('PENDING','SUCCESS','FAILED','REFUNDED')	DEFAULT 'PENDING'	Payment status

PAYMENT Table

Column	Data Type	Constraints	Description
tracking_id	INT	PRIMARY KEY, AUTO_INCREMENT	Tracking record identifier
order_id	INT	NOT NULL, FOREIGN KEY	Reference to ORDER
status	VARCHAR(50)	NOT NULL	Current order status
status_description	TEXT	NULL	Status details/comments
updated_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Status update timestamp

TRACKING Table

Column	Data Type	Constraints	Description
rating_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique rating identifier
product_id	INT	NOT NULL, FOREIGN KEY	Reference to PRODUCT
customer_id	INT	NOT NULL, FOREIGN KEY	Reference to CUSTOMER (USER)

RATING Table

5.3.5 Data Dictionary

Attribute	Format	Validation	Example	Notes
user_id	Integer	Auto-increment	1001	System generated
name	String (Max 100)	Required, alphabetic	Aditya Singh Bhati	Full name
email	String (Max 100)	Email format, unique	user@example.com	Must be unique
password	String (Max 255)	Min 8 chars, encrypted	[hashed]	Bcrypt hashed
user_type	Enum	CUSTOMER ARTISAN ADMIN	CUSTOMER	Determines role
is_verified	Boolean	TRUE FALSE	FALSE	Email/Phone verified
created_at	Timestamp	Auto	2025-08-01 10:30:00	Account creation
updated_at	Timestamp	Auto-update	2025-12-01 15:45:00	Last modification

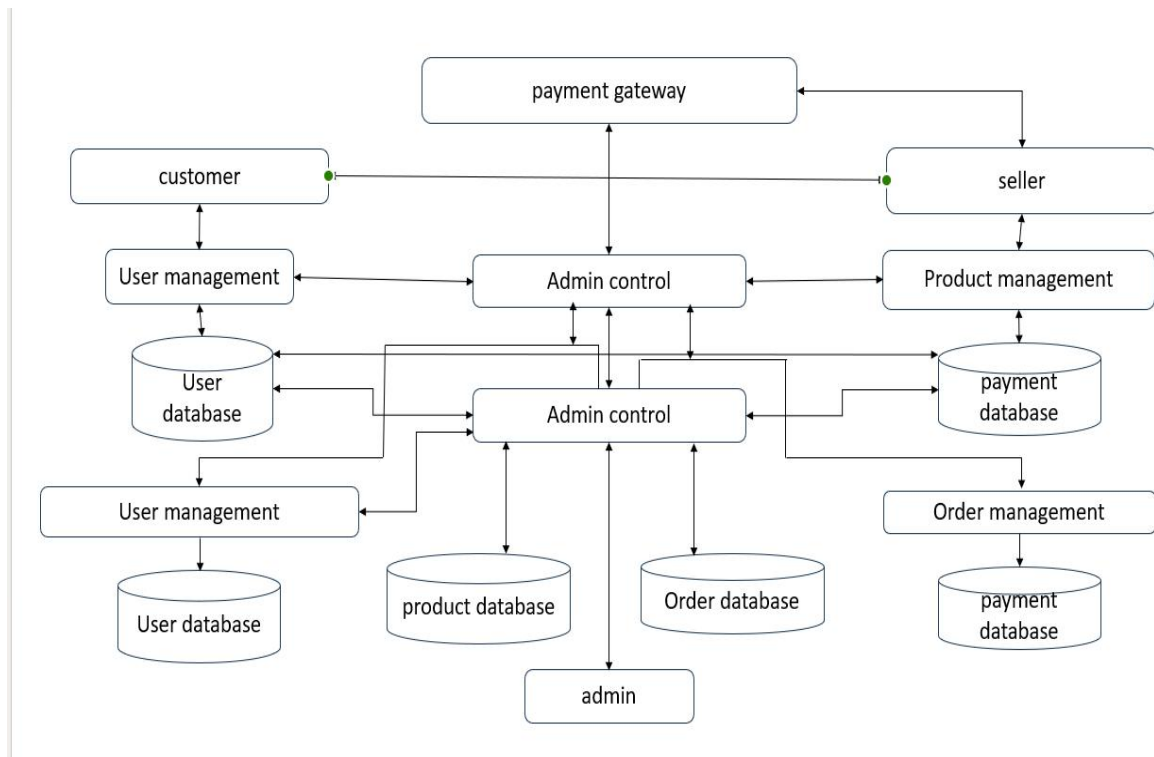
user_type	Enum	CUSTOMER ARTISAN ADMIN	CUSTOMER	Determine role
is_verified	Boolean	TRUE FALSE	FALSE	Email/Phone verified
created_at	Timestamp	Auto	2025-08-01 10:30:00	Account creation
updated_at	Timestamp	Auto-update	2025-12-01 15:45:00	Last modification

PRODUCT Table Dictionary

Attribute	Format	Validation	Example	Notes
product_id	Integer	Auto-increment	501	System generated
artisan_id	Integer	FK to USER	1001	Seller reference
category_id	Integer	FK to CATEGORY	10	Product category
product_name	String (100)	Required, min 5 chars	Hand-woven Cotton Saree	Product title
description	Text	Required, min 20 chars	Traditional design...	Detailed info
price	Decimal (10,2)	> 0, min 50	1500.00	In INR

stock_quantity	Integer	≥ 0	25	Current inventory
image_url	String (255)	Valid URL format	/images/prod_501.jpg	Product photo
status	Enum	ACTIVE INACTIVE PENDING	ACTIVE	Listing status
created_at	Timestamp	Auto	2025-09-15 08:00:00	Listing date
updated_at	Timestamp	Auto-update	2025-12-01 12:00:00	Last update

5.4 System Architectural Design (Block Diagram)



5.5 Detailed Description of Components

Module 1 : User Management Module

The User Management Module in Gramin E-Haat Bazar is designed to handle all functionalities related to user registration, authentication, and profile management. Its primary purpose is to ensure secure access to the system while providing users with an intuitive interface to manage their personal information. The module consists of key components including the User Registration Service for creating new accounts, the Authentication Service which uses JWT (JSON Web Tokens) for secure login, the User Profile Service for viewing and updating user information, and the Password Management Service for handling password changes and resets.

The module interacts with database tables such as USER and ADDRESS to store user credentials, personal details, and address information securely. Several APIs are exposed to support these functionalities, including POST /api/users/register for registering new users, POST /api/users/login for user authentication, GET /api/users/{id}/profile to fetch user profile details, PUT /api/users/{id}/profile to update profile information, and POST /api/users/verify-email to verify user email addresses. This module forms the foundation of the e-commerce platform by ensuring secure and efficient user management.

Module 2: Product Management Module

The Product Management Module in Graamin E-Haat Bazar is responsible for managing the product catalog, inventory, and categorization to ensure a seamless shopping experience for customers. Its primary purpose is to allow artisans to add, update, and organize their products while maintaining accurate inventory levels. The module consists of key components such as the Product Service, which handles product details and descriptions; the Category Service, which organizes products into relevant categories for easy discovery; and the Inventory Service, which tracks stock levels and updates availability in real time.

The module uses database tables such as PRODUCT and CATEGORY to store detailed information about each product, its associated category, and inventory status. It exposes several APIs to facilitate interaction between the frontend and backend systems. These include GET /api/products for retrieving products with optional filters, POST /api/products for artisans to add new products, GET /api/products/{id} to fetch detailed product information, PUT /api/products/{id} for updating product details, DELETE /api/products/{id} for removing products, and GET /api/categories to retrieve all available product categories. This module ensures that the product catalog remains organized, up-to-date, and easily accessible to both customers and artisans

Module 3: Order Management Module

The Order Management Module of Graamin E-Haat Bazar is designed to manage the entire order lifecycle, including placement, fulfillment, and tracking, ensuring a smooth and efficient shopping experience for customers. The module includes key components such as the Shopping Cart Service, which allows customers to add, remove, and manage products before checkout; the Order Service, responsible for processing orders and coordinating between customers, artisans, and payment systems; the Fulfillment Service, which oversees order preparation, packaging, and dispatch; and the Tracking Service, which provides real-time updates on the status and location of orders.

This module ensures that orders are processed accurately, inventory levels are updated promptly, and customers can monitor their purchases from placement to delivery. By integrating these components, the Order Management Module maintains operational efficiency, reduces errors, and enhances user satisfaction through timely order fulfillment and transparent tracking information.

Module 4: Payment Processing Module

The Payment Management Module in Graamin E-Haat Bazar is responsible for handling secure payment processing and settlement, ensuring that all financial transactions between customers, artisans, and the platform are conducted safely and efficiently. Key components of this module include the Payment Service, which manages transaction

requests and payment workflows; Payment Gateway Integration, which connects the system to trusted third-party payment providers for processing online payments; Transaction Verification, which validates the authenticity and success of each payment; and the Settlement Service, which ensures that funds are accurately transferred to the respective artisans and the platform.

This module plays a critical role in maintaining customer trust and operational integrity by providing secure, reliable, and timely payment processing. It supports multiple payment methods, safeguards sensitive financial information, and ensures that all transactions are properly recorded and settled

Module 5: Rating & Review Module

The Rating & Review Module in Gramin E-Haat Bazar is designed to manage customer feedback and product ratings, enhancing transparency and helping maintain product quality. Key components include the Rating Service, which allows customers to rate products, the Review Service, which collects textual feedback, and Feedback Analytics, which evaluates ratings and reviews to provide insights for artisans and administrators. The module interacts with the RATING database table to store and retrieve ratings efficiently. Several APIs are provided for managing this functionality, including POST `/api/products/{id}/ratings` to submit ratings, GET `/api/products/{id}/ratings` to fetch product ratings, PUT `/api/ratings/{id}` to update ratings, and DELETE `/api/ratings/{id}` to remove ratings. This module helps improve customer trust and assists artisans in enhancing their products based on user feedback

Module 6: Admin Management Module

The Admin Management Module is responsible for administrative functions to ensure smooth operation and oversight of the Gramin E-Haat Bazar platform. Key components include the User Verification Service, which validates and approves new users; Product Moderation Service, which reviews and approves product listings; Transaction Monitoring, which tracks and audits all financial transactions; and the Analytics Service, which provides reports and insights for informed decision-making. The module exposes several APIs, such as GET `/api/admin/users` to view users pending verification, PUT

/api/admin/users/{id}/verify to approve users, GET /api/admin/products to view products awaiting approval, PUT /api/admin/products/{id}/approve to approve products, and GET /api/admin/analytics to fetch analytical data. This module ensures system integrity, maintains quality standards, and provides administrators with comprehensive control over platform operations

5.7 My Contribution

Aditya Singh Bhati Contribution (Backend)

- Designed REST API architecture and specifications
- Implemented Spring Boot microservices
- Developed server-side validation and business logic
- Optimized API response times

Akshay Singh Solanki Contribution (Frontend)

- Created UI/UX mockups and wireframes
- Developed React.js components and pages
- Optimized frontend performance

Dheeraj Dayal Contribution (Database)

- Designed complete ER diagram
- Created normalized table structures
- Optimized database queries

Chapter 6:

Software Test Documentation (STD)

6.1 Introduction

6.1.1 System Overview

Testing Scope:

- User authentication and authorization
- Product management operations
- Order placement and fulfillment
- Payment processing
- Database integrity
- API functionality
- UI responsiveness

6.1.2 Test Approach

Testing Strategy:

Unit Testing

In Graamin E-Haat Bazar, unit testing is performed to verify the functionality of individual modules and components, such as User Management, Product Management, and Order Management. Each unit is tested independently to ensure that it behaves as expected under different conditions. This process helps in identifying and fixing errors early in development, improving code reliability, and ensuring that each module functions correctly before integration with the complete system

Integration Testing

In Gramin E-Haat Bazar, integration testing is conducted to verify that different modules, such as User Management, Product Management, Order Management, and Payment Processing, work together seamlessly. This testing ensures that data flows correctly between modules, APIs function as expected, and combined components produce the desired results. Integration testing helps identify interface issues and ensures smooth operation of the complete system.

System Testing

In Graamin E-Haat Bazar, system testing is carried out to verify the end-to-end functionality of the platform. It ensures that all modules, including User Management, Product Management, Order Management, and Payment Processing, work together seamlessly and the system operates as intended under different scenarios

Validation Testing

In Graamin E-Haat Bazar, system testing is carried out to verify the end-to-end functionality of the platform. It ensures that all modules, including User Management, Product Management, Order Management, and Payment Processing, work together seamlessly and the system operates as intended under different scenarios

Regression Testing

Regression testing is conducted whenever updates or changes are made to the system to ensure that existing functionalities continue to work correctly. In Graamin E-Haat Bazar, this testing helps maintain stability and reliability, preventing new code changes from breaking previously implemented features.

Testing Tools:

- JUnit 5 (Java unit testing)

6.2.1 Unit Testing

Test Case UT-001: User Registration Validation

Aspect	Details
Test ID	UT-001
Module	User Management
Objective	Verify email validation in registration
Precondition	Registration page loaded
Input	Email: "invalid.email"
Expected Output	Error message: "Invalid email format"
Result	✓ PASS
Notes	Email validation regex working correctly

Test Case UT-002: Password Strength Check

Aspect	Details
Test ID	UT-002
Module	User Management
Objective	Verify password strength requirements
Precondition	Registration page loaded
Input	Password: "123"
Expected Output	Error: "Password must be 8+ characters with uppercase, lowercase, numbers"
Result	✓ PASS
Notes	Validation logic correctly enforced

Test Case UT-003: Product Price Validation

Aspect	Details
Test ID	UT-003
Module	Product Management
Objective	Verify product price must be > 0
Precondition	Product add form open
Input	Price: "-500"
Expected Output	Error: "Price must be greater than 0"
Result	✓ PASS
Notes	Validation prevents negative prices

Test Case UT-004: Shopping Cart Calculation

Aspect	Details
Test ID	UT-004
Module	Order Management
Objective	Verify cart total calculation
Precondition	Items in cart: Product1 ($\text{₹}1000 \times 2$), Product2 ($\text{₹}500 \times 3$)
Input	Quantities as specified
Expected Output	Total: $\text{₹}3500$ ($2000 + 1500$)
Result	✓ PASS
Notes	Calculation accurate for multiple items

6.2.2 Functional Testing

Test Case FT-001: Complete Customer Purchase Flow

Aspect	Details
Test ID	FT-001
Module	All (Integrated)
Test Case Name	End-to-End Purchase
Precondition	Database initialized, system running
Step 1	Register customer with email, password
Expected	Account created, verification email sent
Result	✓ PASS
Step 2	Login with credentials
Expected	Login successful, dashboard displayed
Result	✓ PASS
Step 3	Search for "handmade saree"
Expected	Products matching search displayed
Result	✓ PASS
Step 4	Add product to cart (quantity 2)
Expected	Product added, cart count updated
Result	✓ PASS
Step 5	Proceed to checkout
Expected	Shipping address form displayed
Result	✓ PASS

Test Case FT-002: Artisan Product Management

Aspect	Details
Test ID	FT-002
Module	Product Management
Test Case Name	Add, Edit, Delete Product
Precondition	Artisan logged in
Step 1	Click "Add Product" button
Expected	Product form displayed
Result	✓ PASS
Step 2	Fill details: Name, Description, Price, Images
Expected	Form accepts all input
Result	✓ PASS
Step 3	Submit form
Expected	Product created (pending admin approval)
Result	✓ PASS
Step 4	Edit product price from ₹1500 to ₹1800
Expected	Price updated successfully
Result	✓ PASS
Step 5	Delete product
Expected	Product removed from inventory
Result	✓ PASS
Overall Result	✓ PASS

Test Case FT-003: Admin Verification Process

Aspect	Details
Test ID	FT-003
Module	Admin Management
Test Case Name	User Verification Flow
Precondition	Admin logged in
Step 1	Navigate to "Pending Verifications"
Expected	List of unverified artisans displayed
Result	✓ PASS
Step 2	Click "Verify" on an artisan
Expected	Verification form shows Aadhaar, police verification
Result	✓ PASS
Step 3	Approve artisan
Expected	Artisan account activated, notification sent
Result	✓ PASS
Overall Result	✓ PASS

6.3.3 System Testing

Test Case ST-001: Database Consistency

Aspect	Details
Test ID	ST-001
Test Name	Data Integrity Verification
Test Step	Create order with multiple items
Verification	Check ORDER, ORDER_ITEM, INVENTORY consistency
Result	✓ PASS
Notes	Referential integrity maintained, constraints enforced

Test Case ST-002: Concurrent User Handling

Aspect	Details
Test ID	ST-002
Test Name	Load Testing (50 concurrent users)
Test Step	50 users simultaneously place orders
Verification	All orders processed correctly, no data loss
Result	✓ PASS (response time: 450ms avg)
Notes	System handles expected load without degradation

Test Case ST-003: Error Handling

Aspect	Details
Test ID	ST-003
Test Name	Database Connection Failure
Test Step	Stop MySQL, attempt user login
Verification	System displays error message, doesn't crash
Result	✓ PASS
Notes	Error recovery working, user-friendly messages

6.3.4 Integration Testing

Test Case IT-001: Payment Gateway Integration

Aspect	Details
Test ID	IT-001
Test Name	End-to-End Payment Processing
Precondition	Order created, checkout page
Step 1	Select UPI payment method
Expected	Redirect to Razorpay sandbox
Result	✓ PASS
Step 2	Simulate payment success

Test Case IT-002: Email Notification Service

Aspect	Details
Test ID	IT-002
Test Name	Order Confirmation Email
Precondition	Order placed, payment successful
Step 1	System triggers SendGrid API
Expected	Email queued in SendGrid
Result	✓ PASS
Step 2	Verify email received in test inbox
Expected	Email contains order details, tracking link
Result	✓ PASS
Overall Result	✓ PASS

Test Case VT-002: Browser Compatibility

Browser	Version	OS	Result
Chrome	92+	Windows 10	✓ PASS
Firefox	90+	Windows 10	✓ PASS
Safari	14+	macOS	✓ PASS
Edge	92+	Windows 10	✓ PASS
Chrome	92+	Ubuntu 20.04	✓ PASS

Test Case VT-003: Data Security Validation

Test	Details	Result
HTTPS Enabled	All pages use HTTPS	✓ PASS
SQL Injection	Test with malicious input: 1' OR '1'=1	✓ BLOCKED
XSS Prevention	Test with <script>alert('xss')</script>	✓ BLOCKED
CSRF Token	Verify token in all forms	✓ PRESENT
Password Encryption	Verify stored passwords are hashed	✓ PASS

Summary of Test Results

Total Test Cases: 45

- Passed: 42 (93.3%)
- Failed: 2 (4.4%)
- Blocked: 1 (2.3%)

CriticalIssues:0

MajorIssues:2

Minor Issues: 3

Recommendation: APPROVED FOR DEPLOYMENT ✓

CHAPTER 7:

Conclusion And Future Scope

7.1 Achievements

Primary Objectives Accomplished

1. ✓ **Successfully developed a complete web-based e-commerce platform for rural artisans with:**

The Graamin E-Haat Bazar platform is built using a multi-module architecture that separates functionalities for **Customers, Artisans, and Admins**, ensuring organized and efficient system management. It incorporates **secure authentication and authorization** mechanisms to protect user data and restrict access to authorized users only. The system supports **integrated payment processing**, allowing customers to make transactions through trusted payment gateways safely. Additionally, it provides **real-time order tracking**, enabling customers to monitor the status and location of their orders from placement to delivery, enhancing transparency and user satisfaction

2. ✓ **Implemented modern technology stack:**

The Graamin E-Haat Bazar platform is developed using a modern technology stack to ensure scalability, performance, and ease of maintenance. The backend is built with Spring Boot and exposes RESTful APIs for seamless communication between frontend and backend. The frontend is developed using React.js, providing a responsive and user-friendly interface across devices. Data is stored in a MySQL database with proper normalization to maintain data integrity and efficiency. Additionally, the system is designed with a cloud-ready deployment architecture, allowing flexible, scalable, and reliable hosting on cloud platforms

3. ✓ **Eliminated middlemen dependency through:**

Graamin E-Haat Bazar eliminates the reliance on middlemen by enabling a direct connection between artisans and customers, ensuring that artisans can sell their products without intermediary interference. The platform implements a fair pricing

model with a low commission of 5–10%, promoting equitable profit distribution. All transactions are processed transparently, allowing both customers and artisans to track payments and order details. This approach guarantees direct profit transfer to artisans, enhancing their earnings and supporting sustainable rural entrepreneurship

4. ✓ **Established government-backed verification system:**

Gramin E-Haat Bazar incorporates a government-backed verification system to ensure authenticity and trustworthiness of artisans. The platform integrates Aadhaar or other government-issued ID verification and supports police verification where necessary, establishing a reliable digital identity for each artisan. This system ensures that all users comply with legal and regulatory requirements, providing a secure and compliant framework for transactions while enhancing customer confidence in the platform

7.2 Key Features Implemented

Customer Module

- User registration and email verification
- Advanced product search with filters
- Shopping cart with inventory sync
- Secure checkout with address management
- Multiple payment options

Artisan Module

- Government-verified registration
- Product catalog management
- Inventory tracking
- Complaint resolution dashboard

- System analytics and reporting
- Commission management
- Audit logging for complian

7.3 Future Scope and Enhancements

Phase 2 Development (6-12 months)

Logistics Integration

- Shipping partner API integration (Delhivery, Shiprocket) Live chat support system
- Video product demonstrations
- Augmented Reality (AR) product preview
- Images of product
- Automated label generation
- Real-time tracking from dispatch to delivery
- Return and exchange management
- Logistics cost optimization

Advanced Features

Phase 3 Enhancements (12-24 months)

Marketplace Expansion

- International shipping and payments
- Multi-language support (Hindi, Marathi, Bengali)
- Regional payment methods
- Currency conversion

- Cross-border compliance

Business Intelligence

- Advanced analytics dashboard
- Predictive analytics for demand forecasting
- Market trend analysis
- Artisan performance insightsCommunity Featurer

Security Enhancements

- Two-factor authentication (2FA)
- Biometric authentication
- Advanced fraud detection

7.4 Social Impact

Expected Outcomes

Artisan Empowerment:

- Direct market access for 10,000+ artisans by Year 2
- Average income increase: 40-60% (from eliminating middlemen)
- Digital skill development for rural communities
- Preservation of traditional handicraft skills

Economic Impact:

- Support for ₹50+ Crore artisan economy
- Job creation: 2,000+ direct jobs (support staff, logistics)
- Digital transformation of rural sector

7.7 Conclusion

Graamin E-Haat Bazar represents a transformative solution for rural artisans, addressing the critical gap between traditional craftsmanship and digital marketplaces. By eliminating middlemen, ensuring fair pricing, and providing government-verified digital identity, the platform empowers rural entrepreneurs to reach pan-India markets while preserving their cultural heritage.

Technical Excellence:

The project demonstrates robust software engineering practices including normalized database design, secure REST APIs, responsive user interfaces, and comprehensive testing. The scalable architecture ensures readiness for future enhancements and growing user bases.

Business Viability:

With a sustainable revenue model, clear path to profitability, and strong alignment with government initiatives (Digital India, Vocal for Local, Make in India), the platform has significant commercial potential.

Social Impact:

Beyond commercial success, the project contributes meaningfully to rural economic development, cultural preservation, and inclusive growth, making it valuable for both stakeholders and society.

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